11 *Putting Things Together*

Definition 11.1: Graph.
A *graph* G is determined by a *finite* set of *vertices* $V(G)$ and a set of *edges* $E(G)$ satisfying $E(G) \subseteq \{W \mid W \subseteq V(G) \wedge |W| = 2\}$.

Definition 11.2: Path.
Let G be a graph. A *"walk"* along G is meant to denote a possible way of visiting the vertices of G by traveling along the edges. Given a natural number $k \in \mathbb{N}$, we define a *walk of length k on G* to be a function $f : (k+1) \to V(G)$ such that, for each $i < k$, we have $\{f(i), f(i+1)\} \in E(G)$.[1] A *path* is a walk that does not repeat any nodes; more formally, a *path of length k on G* is an injective walk of length k on G.[2]

Definition 11.3: Cycle.
Given $k \in \mathbb{N}$, a *circuit of length k* on a graph G is a walk $w : k+1 \to V(G)$ that starts and ends at the same place, meaning $w(0) = w(k)$. If a circuit does not repeat any nodes other than the first and last ones,[3] then we call this circuit a *cycle*.

Theorem 11.1: Handshake Lemma (v. 1).
For any graph G, we have $\sum_{v \in V(G)} \mathtt{deg}_G(v) = 2 \cdot |E(G)|$.

Theorem 11.2: Handshake Lemma (v. 2).
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Theorem 11.3: Handshake Lemma (v. 3).
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[1] This means that consecutive nodes $f(i)$ and $f(i+1)$ along the walk must be connected by an edge in G.

[2] Notice the *length* of a walk or path is determined by the number of *edges* involved, not the number of *vertices*.

[3] … meaning the walk is essentially a *path* except for the fact that the first and last nodes are the same…